

Blood Bank Inventory & Emergency Donor Matcher — UML Component Diagram

Presentation Layer



Request Submission Interface

internal API

Business Layer

«component»
Emergency Request Manager
Validate request, urgency & requester

Matching Interface

service call

SMS Notification Interface

notification API

Inventory Availability Interface

service call

«component»
Inventory Management
Stock updates, availability & expiry flags

«component»
Compatibility & Donor Matching
ABO/Rh(D) verification + donor radius matching

«component»
Notification Service
Emergency SMS alerts

Inventory Data Interface

database queries

Donor Data Interface

database queries

Data Layer

«component»
Blood Bank Data Repository
Inventory, donor and emergency-request data

UML interface notation: Ball = provided interface | Socket = required interface | Solid line = assembly connector